Converting Mechanical Energy to Electrical Energy via Piezoelectric/Electromagnetic Panels

Problem

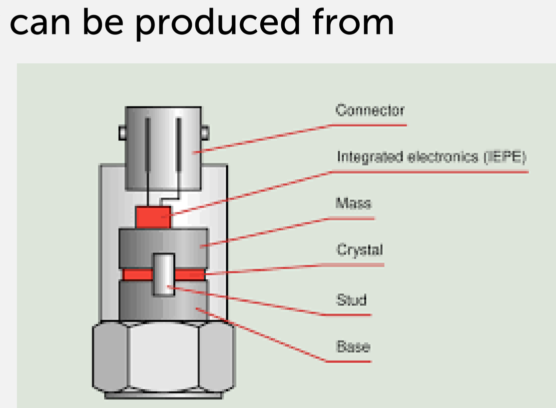
Over the past 20 years, the energy needs of major cities, such as New York City and Los Angeles, have increased gradually and are expected to continue rising. Certain percentages of total funding are dedicated to meet electrical quotas, and as these energy demands grow, measures should be taken to maximize total energy production, while minimizing cost and allowing funding to be reallocated.

Purpose

An increasing electricity demand in these larger cities should result in higher bills for the residents in this area, something that can and should be prevented so as not to lower the affordability of these communities. Given the limited space in these cities, methods for energy production should be prioritized that make efficient use of available space and/or generate energy in areas that previously could not. These solutions should also remain affordable, so as not to substantially raise the spending governments already put into infrastructure.

Background

Piezoelectric transducers allow for the direct conversion of mechanical energy, or pressure, to an electrical voltage through the use of piezoelectric crystals. Using the equation Q=dF, where Q is the charge, d is the charge sensitivity, or piezoelectric constant, and F is the applied force in Newtons, an estimate can be made for total energy production. When incorporated with various other means for energy production, significant amounts of energy can be produced from moderate amounts of force.



Piezoelectric Transducer

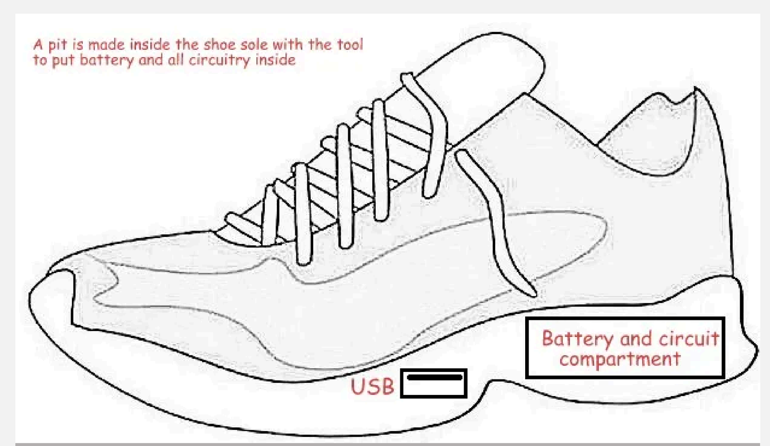
Pavegen's kinetic tile is the perfect example, using a combination of both piezoelectric transducers and electromagnetic induction to produce large amounts of energy with just a single step. This makes them the perfect framework to build off of to achieve the final goal.


Pavegen's Kinetic Tile

Electromagnetic induction is the most complex factor for Pavegen's tile, but it can be simplified into the equation $e = N\, d\Phi/dt$, with e being the instantaneous induced voltage, N being the number of turns in the wire, Φ being the magnetic flux in Webers, and t being time in seconds. Basic induction can be reached by simply placing a magnet between a coiled copper wire.

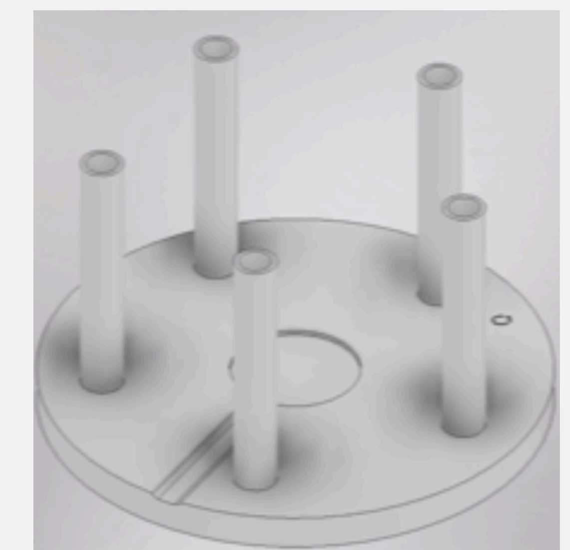
Demonstrator

The initial concept for the demonstrator came from Kavish Laxkar, an electrical engineer, who incorporated piezoelectric transducers to generate energy via a USB-A port. He did this in the form of a shoe, placing the transducer underneath the sole, along with other components such as the wiring and the battery, so the person's footsteps compress the transducer and generate electrical energy. This was almost exactly what I was looking for, but it failed to show the energy generated from the electromagnetism, so I couldn't directly use this concept for my project.



Laxkar's Concept Design for His Electricity Producing Footware

I ended up deciding on a model that allows for the hand compression of a transducer, and a magnet in the center that goes through copper wiring, giving the electromagnetic energy production. The center circle is meant to house the transducer, the line going to the edge allowing for its wires to leave, while the cylinders on the outside store the springs, keeping the top from compressing fully. The top section is simply a cylindrical magnet and prongs meant to fit on top of the springs in the bottom half

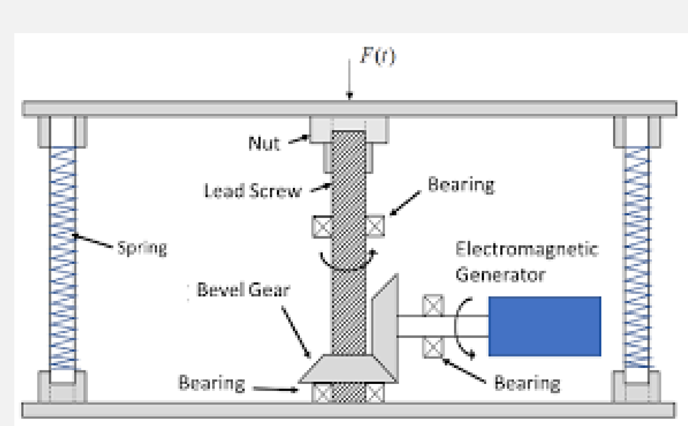

Bottom Half of the Demonstrator Model

Prototype

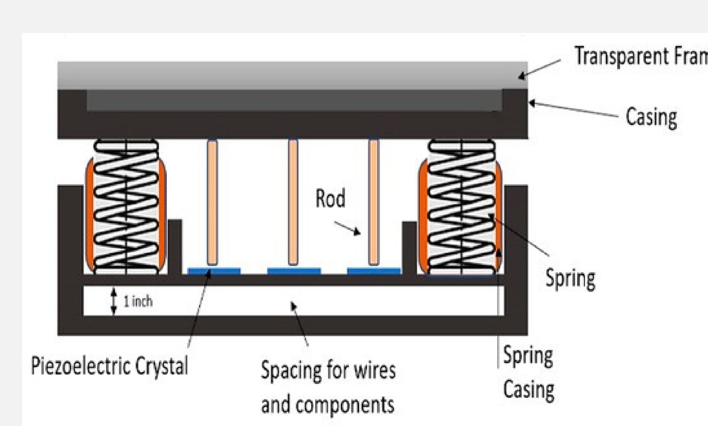
The prototype took inspiration from two separate designs:

The first being a model allowing for a screw to move vertically, with the threading on the screw allowing for the rotation of two bevel gears and the activation of an electromagnetic generator, allowing for the production of electrical energy.
The second design requires a collection of rods moving vertically and compressing piezoelectric transducers. As these transducers are compressed, the crystals within them are squeezed, and as discussed in the background section, allow for the production of electrical energy.

Both of these designs have relatively similar bases and shapes to what I was looking for in the demonstrator, and given their relative simplicity, their designs should be easy to combine for my final prototype.

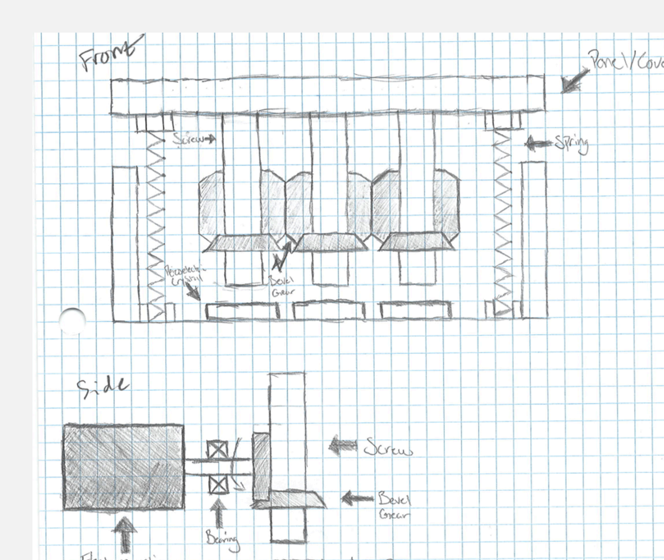


Screw Based Design with Electromagnetic Generator



Rod Based Design with Piezoelectric Crystals

This final design, as shown in the sketch, takes the best parts of each of these models, the electromagnetic generator demonstrated in the first, and the piezoelectric crystals that I wanted to incorporate from the second. With a combination of eight sides (24 total screws in total) and one electromagnetic generator in the center, this was meant to be the final design for the prototype.


Prototype Design Sketch

Data and Results

Given that the prototype was not able to be constructed by the end of the project, the data from the demonstrator will be used to display the results. The overall purpose of the demonstrator, as stated previously, was to show that through the usage of both piezoelectricity and electromagnetic conduction, electrical energy could be produced through the application of mechanical energy. After the application of a ten gram object (a book) to the top of the demonstrator, the theoretically calculated value for electrical energy produced 3.1 Watts, but the experimental value that was achieved was only around 0.78 Watts, a questionable 25.2% yeild. This leads me to believe there were possible errors in either the calculation of the theoretical value, not placing enough limiters on energy gained or lost, or experimentation errors in how the data was collected.

Conclusion

Although the primary objective of this project was not achieved, the creation of a prototype panel that both produced electricity and was cheap to manufacture, the basic concepts I was looking for can be determined through the demonstrator and the research I conducted. The initial question, can electrical energy be produced through the use of mechanical energy on piezoelectric transducers, is shown to be true due to the data collected through the demonstrator. The expected energy production was drastically higher than the achieved values, but this is most likely due to errors in data collection, and the lack of knowledge about the transducers I was using due to them being bought from Amazon and the seller not disclosing the specific piezoelectric material within the transducer. The overall lack of cost that was required to produce this energy, though, is a slight benefit, showing that this technology could have a place in a city's energy production infrastructure in the near future, keeping in mind that this is already done in European and Asian countries such as the United Kingdom and Japan.

Future Iterations

Given that this project was not able to be physically created after designing the prototype, the most logical next step would be to follow through with its construction. There were some additional things that could have been added to make this concept more useful for use cases outside of just energy projection. One of which was the addition of an Arduino within the panel to transfer information to an external device, detailing how much energy was produced at a time and how many times energy was produced. This would be great for data collection, as it shows areas with the most traffic, and possibly shows average weight data in an area, due to heavier people producing more energy than lighter people. I would also like to move away from the piezoelectric transducers I purchased and try to create my own from scratch, giving me more influence on the variables I wish to include when creating my panels, compared to having to orient my panels around the transducers that are available to be purchased

References

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